

THE SMART HYDROPONIC FARMING

Varsha Patil

(Professor)

AISSMS Institute of Information Technology

Pune, India

varsha.patil@aisssmsioit.org

Shreya Chougule , Srutika Borge , Akshada Bhujbal , Mrunali Kale , Saniya Desai , Prathamesh Bansode , Mahesh Davhare , Tushar Chougule.

(Undergraduate Scholars),

Department of Electronics and Telecommunications.

1.ABSTRACT

The Smart Hydroponic Farming System with LED Grow Lights is an IoT-based project developed to enhance modern agriculture through automation and resource optimization. This system allows plants to grow without soil, using a nutrient-rich water solution that ensures efficient use of water and nutrients. An ESP32 microcontroller monitors key environmental parameters such as temperature, humidity, pH, and water level, while live data is displayed on the Blynk IoT application for real-time observation. A unique feature of this project is the use of four different LED light colors Red, Green, UV, and White to study their individual effects on the growth of lettuce plants. This comparative and automated system promotes smart, sustainable, and space-efficient farming, offering an effective solution for urban agriculture and future smart farming technologies.

Keywords: Hydroponic Farming, Led Grow Lights, IoT, ESP32, Smart Farming Technology.

2.INTRODUCTION

Agricultural production today faces critical challenges due to diminishing arable land, excessive water consumption,

and unpredictable climatic variations. These constraints have accelerated the global shift toward controlled-environment agriculture. Hydroponic farming, a soil-less cultivation technique, has emerged as an effective alternative, allowing plants to grow in a nutrient-enriched aqueous medium. This approach provides precise control over growth parameters, enhances resource efficiency, and supports year-round production independent of soil fertility or weather conditions. The present study, titled "Smart Hydroponic Farming System with LED Grow Lights," integrates Internet of Things (IoT) technologies to optimize and automate hydroponic cultivation processes. The system continuously monitors and regulates environmental factors such as temperature, humidity, pH level, and water level through sensor networks interfaced with an ESP32 microcontroller. By incorporating LED grow lights of varying spectrums red, green, ultraviolet (UV), and white the system facilitates experimental analysis of light wavelength effects on lettuce growth performance. This enables a better understanding of how different spectral compositions influence photosynthetic activity and overall plant development, contributing to improved efficiency and sustainability in agricultural practices [1]. Conventional hydroponic setups often rely on manual supervision and lack systematic evaluation of lighting impacts, resulting in inconsistent growth patterns and inefficient use of resources. The proposed system addresses these limitations by automating environmental control and enabling real-time data acquisition through IoT platforms such as the Blynk application and optionally, Thing Speak for data analytics and visualization. This integration ensures minimal human intervention while maintaining optimal conditions across all growth compartments. The hardware configuration includes an ESP32 microcontroller, DHT11 sensor for temperature and humidity measurement, pH sensor, and ultrasonic sensor for water level detection. Each of the four hydroponic chambers is illuminated with distinct LED light sources—red, green, UV, and white—to facilitate comparative photobiological studies. A transparent hydroponic tank with an air pump maintains oxygenation, and the overall system is powered through dual supply lines: 5V DC for control of electronics and 12V DC for LED operations. The compact setup is suitable for indoor, rooftop, and greenhouse farming environments.[2]

3.METHODOLOGY

Materials Preparation and System Layout
Gather all components: 4 DHT11 sensors, 1 pH sensor, 1 humidity sensor, 1 ESP32, motor controller, 12V water pump, water level sensor, UV/red/green/white LEDs, hydroponic nutrients, four lettuce seedlings, and compartment containers. Design a base frame or hydroponic tray divided into 4 equal compartments, each isolated from light leakage from the others for consistent LED effects.

Colour	Name	Wavelength(nm)	Voltage drops(v)
Red	Red	610-760	1.6-2.0
Green	Green	500-570	1.9-4.0
White	White	395-530	3.0-5.0
UV	UV	<400	3.1-4.4

Table: 3.1 led name and wavelength

Mount the LEDs directly above each compartment, about 20-25 cm above plant canopy for even light distribution

Electronics and Sensor Integration
Wire DHT11 sensors: Place one in each compartment to measure temperature and humidity. pH sensor: Submerge in the central nutrient reservoir to monitor pH levels. Water level sensor: Place in the common water tank or main nutrient solution to detect level changes. ESP32 connections: Connect all sensors and control wires to digital/analog pins on ESP32 according to your schematic. Use proper voltage regulators/shifting if required.

Plumbing and Motor Setup
Route water lines: Connect the 12V water pump to a distribution line that feeds each compartment equally (drip or NFT method). Link the pump to the motor controller and then to ESP32, so automated watering can be triggered according to sensor or schedule logic. Ensure each compartment has a drain or overflow to

maintain root zone oxygenation.

LED System Installation: Install each LED (UV, red, green, white) securely above its corresponding compartment. For consistent results, standardize distance from LED to plant and intensity (use same power supply/current-limiting circuit for all LEDs). Use reflective material inside compartments to minimize light loss.

Firmware and IoT Monitoring Program
Program the ESP32 to: Read all sensor data (temperature, humidity, pH, water level). Control motor (pump ON/OFF) based on a set schedule or soil/solution moisture data. Log sensor readings to Blynk or other IoT platforms for real-time monitoring. Allow manual/automatic switching between LEDs if testing protocols require cycling conditions.

Testing and Documentation
Plant one lettuce seedling per compartment. Add equal nutrient solution to all compartments or ensure common water circulation. Set LEDs for a fixed light period (e.g., 14-16 hours/day) for each compartment. Monitor and record plant growth over 2-4 week capture photos, take weekly plant height/mass measurements, and log environmental sensor data. Compare lettuce growth (height, leaf size, color, root mass) under each LED type at the end of the experiment.

Circuit Diagram

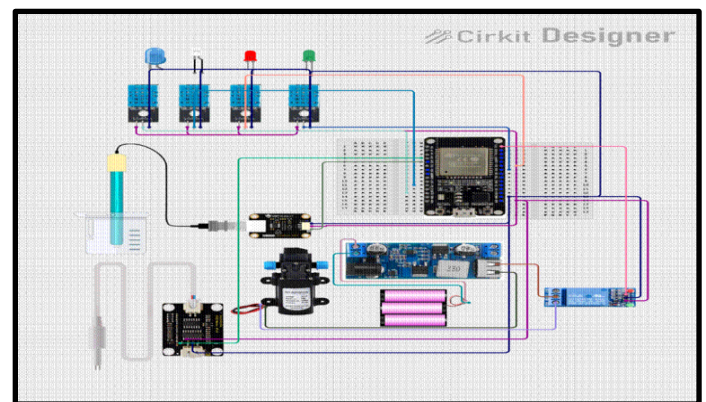
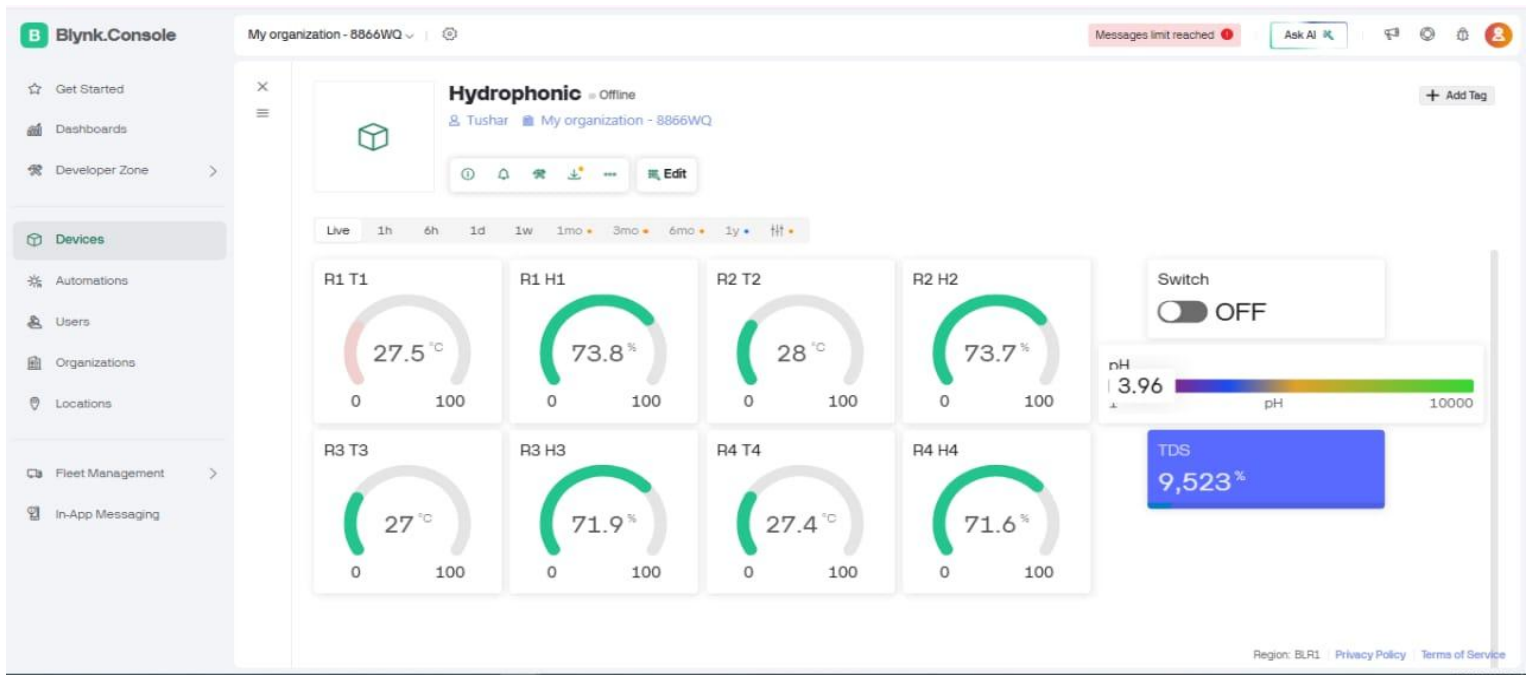


Fig.3.1 circuit Diagram

4.Result



The developed Smart Hydroponic Farming System with LED Grow Lights successfully monitored and controlled key parameters such as temperature, humidity, pH, and water level using the ESP32 microcontroller and Blynk IoT app.

The system automatically maintained suitable growing conditions and allowed real-time monitoring through a smartphone.

Experiments were conducted using Red, Green, UV, and White LED lights on lettuce plants to study their growth patterns.

The ESP 32 controller and smartphone were programmed with the Blynk IoT application, and connectivity between the ESP 32 controller and the smartphone was established. The shortcut of the final Blynk IoT app.

The experiments proved that monitoring of the hydroponics system was successfully carried out. For two weeks, daily statistics data from the Blynk app were gathered at the same time. Most of the results, except for temperature, were constant, as shown in Table 2, because the system compensates for changes in water, salt, or nutrition liquid. This system can be used with any plant that can be grown hydroponically, and the programming can be modified for various parameters, including the pH and salt percentage that suit the plant's needs. .

Day	EC(pp m)	pH	Temp(celcius)
1	1780	6.8	20.5
2	1778	6.8	20.5
3	1781	6.9	21
4	1778	6.8	21.5
5	1779	6.9	22
6	1777	6.8	21.5
7	1782	6.7	22
8	1778	6.5	21.5
9	1778	6.2	20.5
10	1782	6.3	20.9
11	1777	6.7	21.2

Table:4.2 observations from blynk.



Fig.4.3Before Plantation with grow lights.

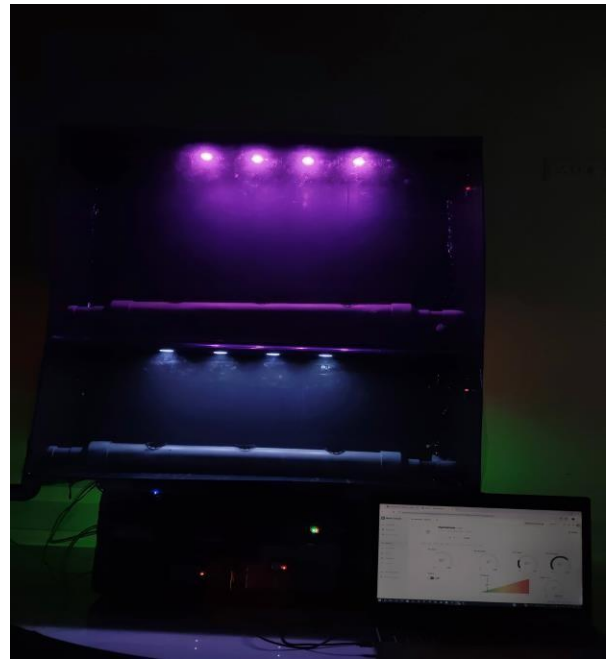


Fig.4.4Before Plantation with grow lights.



Fig.4.5After Plantation Initial stage

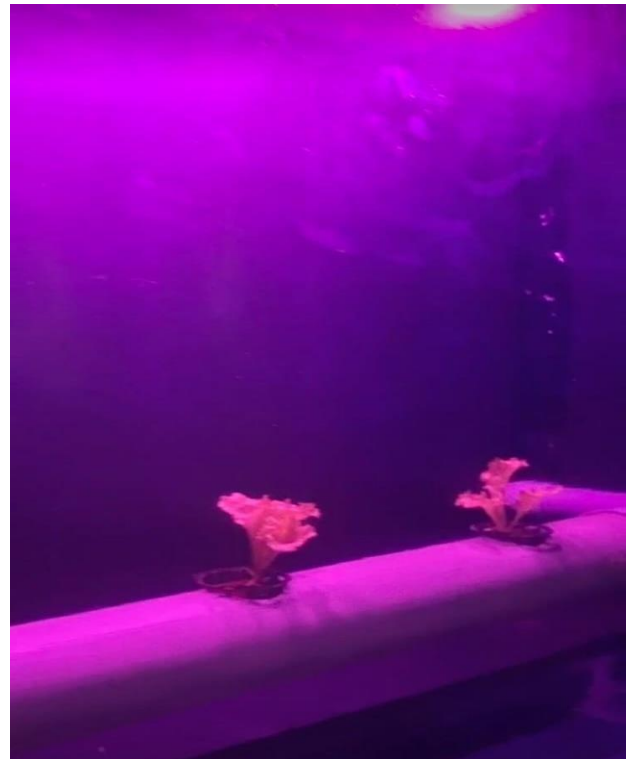


Fig.4.6After Plantation Initial stage

6.Reference

5.Conclusion

The project “Smart Hydroponic Farming System with LED Grow Lights” successfully demonstrated an efficient and sustainable approach to modern agriculture. The system integrated IoT technology with hydroponic cultivation to automatically monitor and control essential parameters such as temperature, humidity, pH, and water level. This automation minimized manual effort and ensured optimal conditions for plant growth. [15]

Experimental results showed that different LED light colors had significant effects on the growth of lettuce plants. Among them, the Red LED promoted the fastest and healthiest growth due to its strong influence on photosynthesis, while the White LED provided balanced, steady development similar to natural sunlight.

The system effectively reduced water usage by up to 90%, eliminated the need for soil, and operated with low power consumption, making it ideal for urban, indoor, and rooftop farming.

Overall, this project proved that integrating smart IoT-based monitoring with hydroponic systems can greatly enhance agricultural productivity, efficiency, and sustainability. It provides a practical step toward future automated and eco-friendly farming solutions. [23]

Future Work:

- Integration with AI for automated nutrient and light adjustment.
- Mobile app-based real-time monitoring and alerts.
- Expansion to more types of crops.
- Solar-powered hydroponic systems for sustainability.
- Data-driven analysis of plant growth across multiple conditions.

- [1]Smart Control Models Used for Nutrient Management in Hydroponic Crops: A Systematic Review – Pablo CatotaOcapana, Cesar MinayaAndino, Paul Astudillo, Diego Pichoasamin
- 2 Elucidation of Intelligent Classification Framework for Hydroponic Lettuce Deficiency Using Enhanced Optimization Strategy – Mustufa Haider Abidi, Sanjay Chintakindi, Ateekh Ur Rehman, Muneer Khan Mohammed
- 3 Recent Developments of the Internet of Things in Agriculture: A Survey – Vippon Preet Kour, Sakshi Arora
- 4 Design of a Smart Hydroponics Monitoring System Using ESP32 Microcontroller and IoT – Anees Abu Sneineh, Arafat A.A. Shabaneh
- 5 Smart Hydroponic Greenhouse: Internet of Things and Soilless Farming – Sara Bardi, Claudio E. Palazzi
- [6]Light Emitting Diode (LED) Lights for the Improvement of Plant Performance and Production: A Comprehensive Review – Saikat Sena, Soni Kumari, Vijay Kumar, Azamal Husen
- [7]A Smart Energy Monitoring System Using ESP32 Microcontroller – Hala Jarallah ElKhozondar et al.
- [8]Advances in Sensing and Monitoring Technologies for Precision Hydroponic Cultivation – Shamim Ahamed et al.
- [9]Smart Farm for Hydroponic Cultivation Using Integrated Renewable Energy Systems – Hoyoung Kim, Inwoo Kim, JeongWoo Seo, Junghyuk Ko
- [10]Smart Hydroponic Farming System Integrated with LED Grow Lights – Primadiyanti Nirbita, Kah-Yoong Chan, Gregory Soon How Thien, Chu-Liang Lee
- [11]Indoor Hydroponic System Using IoT-Based LED (LOTUS) – Wiedjaja Atmadja, Ivan Alexander, Satrio Dewanto, Adela Cahya Nugraha, Stefan Gokparulian
- [12]The Effect of LED Lighting on Lettuce Growth in a Vertical IoT-Based Indoor Hydroponic System – Sanya Kuankid, Apinan Aurasopon
- 13 Effect of Light Emitting Diode (LED) Spectra on Plant Growth – Nur Syahirah Talib, Diyana Jamaludin, Nur Sakinah Abdul Malek
- 14 IoT-Based Hydroponics System – Mwaniki Grace Waigumo, Ruth Wanjiru Mwaniki, Mustafa Mansoor

- 15 Smart Hydroponic Systems: Optimizing Nutrient Levels with IoT Connectivity – Pradnya Vishram Kulkarni, Dr. Vinaya Gohokar, Mr. Kunal Kulkarni
- 16 Enhancing Hydroponics Efficiency Using IoT-Based Automated Monitoring and Sustainable Crop Production System – Dr. Prashant Nitnaware, Monicka Nitin Jagtap
- 17 Hydroponic Smart Farming with IoT – Dr. Prathima V.R, Parvathi C, Dr. Nirmala S. Guptha
- 18 Role of LED Light Illumination on the Growth of Plants in Greenhouse Farming-Hydroponics: IoT Technology – Ravindra B. Malabadi et al.
- [19]Hydroponics: Current Trends in Sustainable Crop Production – Ganapathy Rajaseger, Kit Lun Chan, Kay Yee Tan, Shan Ramasamy, Mar Cho Khin, Anburaj Amaladoss, Patel Kadamb Haribhai
- [20]Hydroponics with LED Lights in UAE – Nexsel Tech Team
- [21]Contemporary Smart Hydroponics Systems: Taxonomy, Enabling Technologies, and Challenges – Abeer Ahmed et al.
- [22]Recent Developments and Status of LED Technology in Controlled Environment Agriculture – Bonga L. Ngcobo, Isa Bertling
- 23 Review of Grow-Light Hydroponics, Irrigation, and Smart Farming – R. Tharun, K. G. Padmasine
- 24 LED Illumination for High-Quality Crop Growth in Protected Cropping – Md M. Rahman et al.
- 25 The Effect of Using LED Lighting in Hydroponic Crop Growth – Luechai Promratrak.